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Talent Intelligence AI

A 100% Offline, Two-Tower Hybrid Retrieval Ranking Engine

Team Name: [Your Team Name]

Track: Track 1 — The Data & AI Challenge

The Problem (Vocabulary Mismatch)

Standard AI ranking systems are flawed.

- Most vector-based AI models use "Dense Embeddings".
- **The Flaw:** If a JD requires "PyTorch", a semantic model might highly rank someone who only knows "TensorFlow".
- **The Solution:** We needed a way to perfectly balance deep meaning with exact keyword matching—without using external APIs.

Our Solution: True Hybrid Search

We built a Silicon Valley-grade "Two-Tower" Retrieval Engine:

- 🏰 **Tower 1: Dense Scoring (70%)**
Uses all-MiniLM-L6-v2 to understand deep semantic context.
- ⚡ **Tower 2: Sparse Scoring (30%)**
Uses TF-IDF vectorizer to perform exact lexical keyword matching on the fly.

Result: Never misses a mandatory keyword, but still understands nuanced career trajectories.

The 3-Layer Architecture

- **Layer 1 (Hybrid Semantic Match):** Blends Sparse + Dense scores.
- **Layer 2 (Structured Features):** Evaluates Years of Experience and Core AI Skills.
- **Layer 3 (Behavioral Signals):** Analyzes Redrob metrics (Response Rate, GitHub).
 - *Penalty Mechanism:* Inactive candidates are mathematically pushed out of the Elite Shortlist.

The Enterprise Full-Stack Experience

- **Backend:** Blazing fast **FastAPI** Python architecture.
- **Frontend:** Modern **React + Vite** SPA.
- **Aesthetics:** Stunning glassmorphic UI via **Tailwind CSS v4**.
- **Analytics Dashboard:** Integrated **Recharts** to dynamically generate Score Distribution curves and Scatter plots to instantly spot "Hidden Gems".

Rule Compliance & Security

100% Offline & Hackathon Compliant

- **Zero Network Calls:** No OpenAI/Gemini APIs. Models are fully cached locally.
- **Strict Format Output:** We successfully output the exact required `submission.csv`.
- **Open Source:** Full code, dependencies, and architecture documentation pushed to GitHub.

Thank You

Ready to revolutionize recruitment.

- **GitHub:** <https://github.com/Himasri2706/india-runs-ranker>
- **Questions?**